

WEEK 3 REPORT: Financial Data Import, Transformation & Time Intelligence

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1. Title

Shopify Stock Market Analysis Using Power BI

Dataset: Shopify Stock Data

Tool Used: Microsoft Power BI

Project: Week 3 – Financial Data Import, Transformation & Time Intelligence

2. Introduction

This project focuses on analyzing Shopify stock market data using Microsoft Power BI. The dataset contains historical stock information such as opening price, closing price, highest price, lowest price, adjusted closing price and trading volume.

Power BI was used to import and prepare the stock data, create a dedicated Date Dimension table, establish relationships between tables, develop DAX measures and build an interactive dashboard.

The main purpose of the project is to understand how Shopify's stock prices and trading activity change over time.

3. Objectives

The main objectives of this project are:

- To import and prepare Shopify stock data in Power BI.

- To create a dedicated Dim_Date table using DAX.
- To establish a one-to-many relationship between the Date table and stock data.
- To create financial measures using DAX.
- To implement time-intelligence functions such as DATEADD(), TOTALYTD() and SAMEPERIODLASTYEAR().
- To create Year, Quarter, Month and Day hierarchies.
- To visualize Shopify stock performance through an interactive dashboard.

4. Dataset Description

The dataset used for this project is **Shopify Stock Data**.

The main columns used in the analysis are:

Column	Description
date	Date of the stock data
open	Opening stock price
high	Highest stock price
low	Lowest stock price
close	Closing stock price
adj_close	Adjusted closing price
volume	Trading volume

4. Data Import and Preparation

The Shopify stock dataset was imported into Power BI Desktop.

After importing the data, the main table was renamed to:

Stock_Data

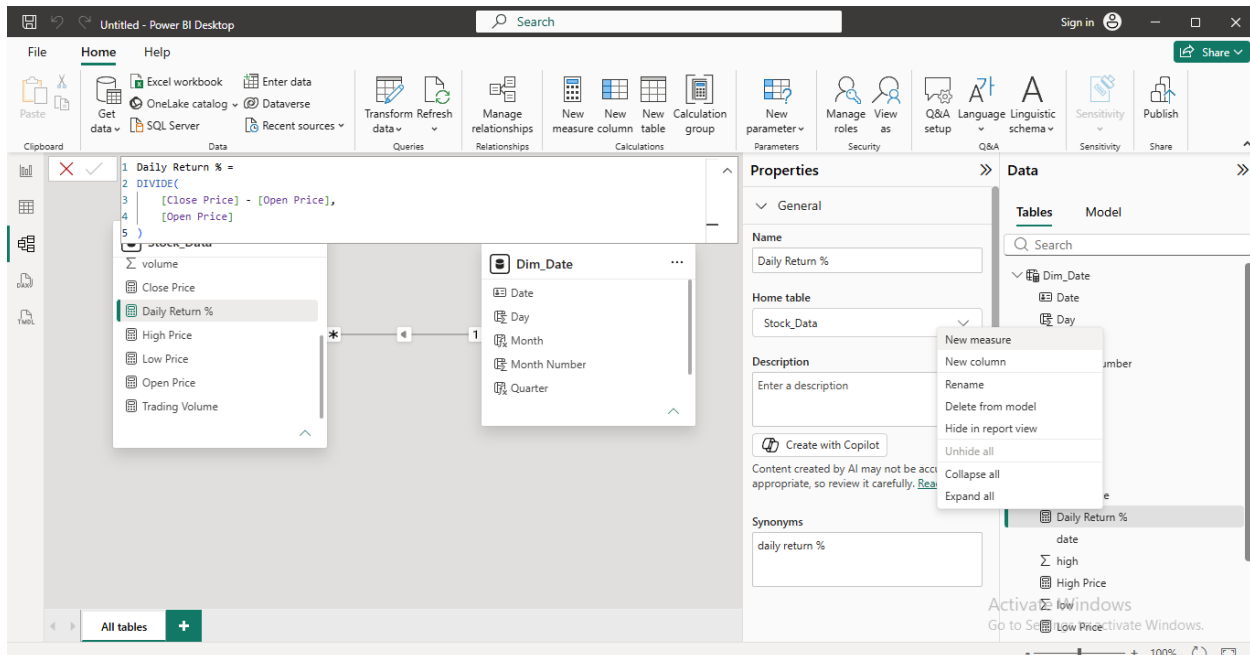
The data contains stock price and trading activity information required for financial analysis.

The important fields used in the project were date, open, high, low, close and volume.

5. Date Dimension

A separate Date Dimension table named **Dim_Date** was created using the DAX CALENDAR() function.

The Date Dimension provides a structured way to perform time-based analysis.



6. Data Model and Relationship

A relationship was created between the Date Dimension and the stock data table.

The relationship is:

Dim_Date[Date] → Stock_Data[date]

The relationship uses a **one-to-many (1:*)** cardinality.

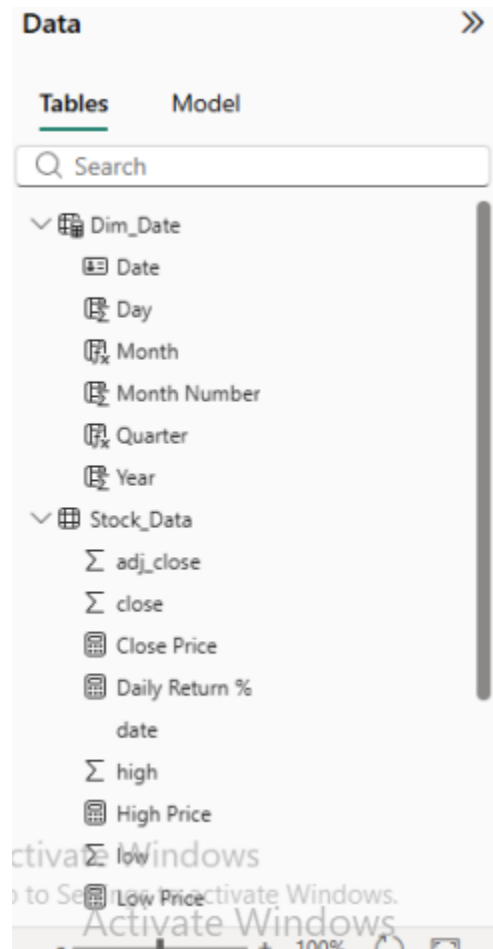
- Dim_Date is on the **one** side.
- Stock_Data is on the **many** side.

This relationship allows the Date Dimension to filter and organize the stock data for time-based analysis.

7. Financial DAX Measures

Several DAX measures were created to analyze the financial data.

This measure calculates the percentage change between the opening and closing prices.



8. Time Intelligence

Time-intelligence functions were used to perform date-based comparisons and analysis.

8.1 Previous Day Close

The DATEADD() function was used to retrieve the closing price for the previous date.

```
Previous Day Close =  
CALCULATE(  
    [Close Price],  
    DATEADD(Dim_Date[Date], -1, DAY)  
)
```

8.2 YTD Close

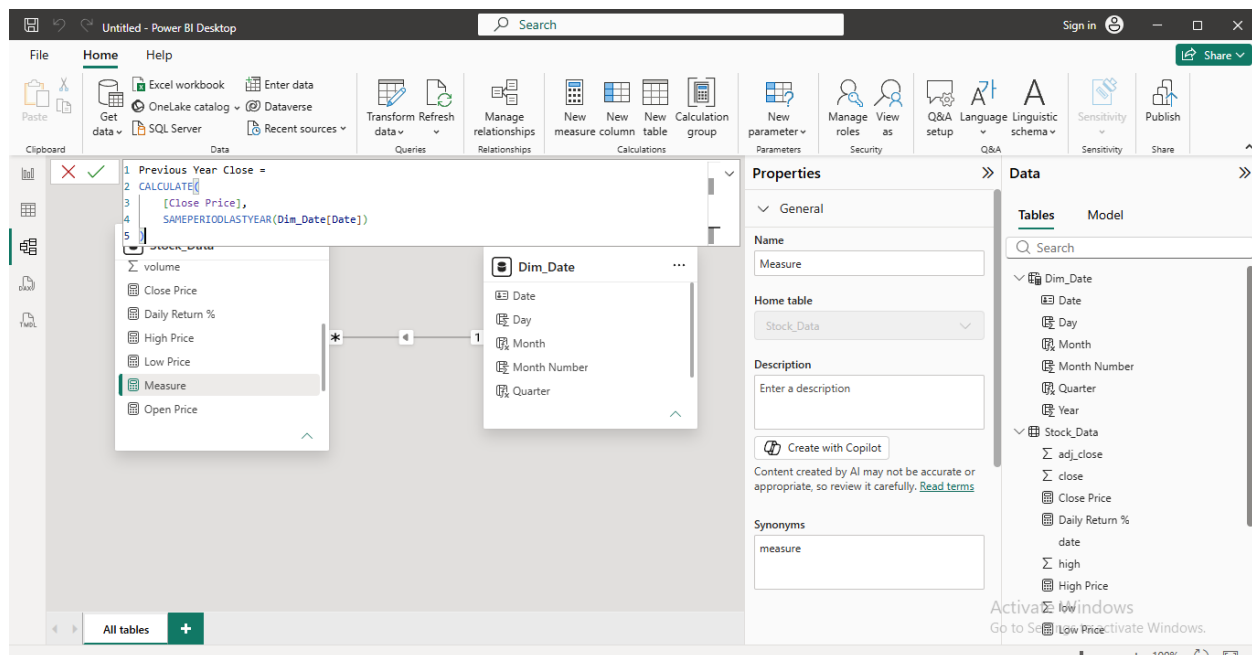
The TOTALYTD() function was used to calculate the year-to-date closing price.

```
YTD Close =  
TOTALYTD(  
    [Close Price],  
    Dim_Date[Date]  
)
```

8.3 Previous Year Close

The SAMEPERIODLASTYEAR() function was used to compare the current period with the corresponding period of the previous year.

```
Previous Year Close =  
CALCULATE(  
    [Close Price],  
    SAMEPERIODLASTYEAR(Dim_Date[Date]))
```



9. Dashboard

A Power BI dashboard titled **SHOPIFY STOCK MARKET ANALYSIS** was created to provide a visual summary of the stock data.

The dashboard contains five KPI cards:

1. Open Price
2. Close Price
3. High Price
4. Low Price
5. Trading Volume

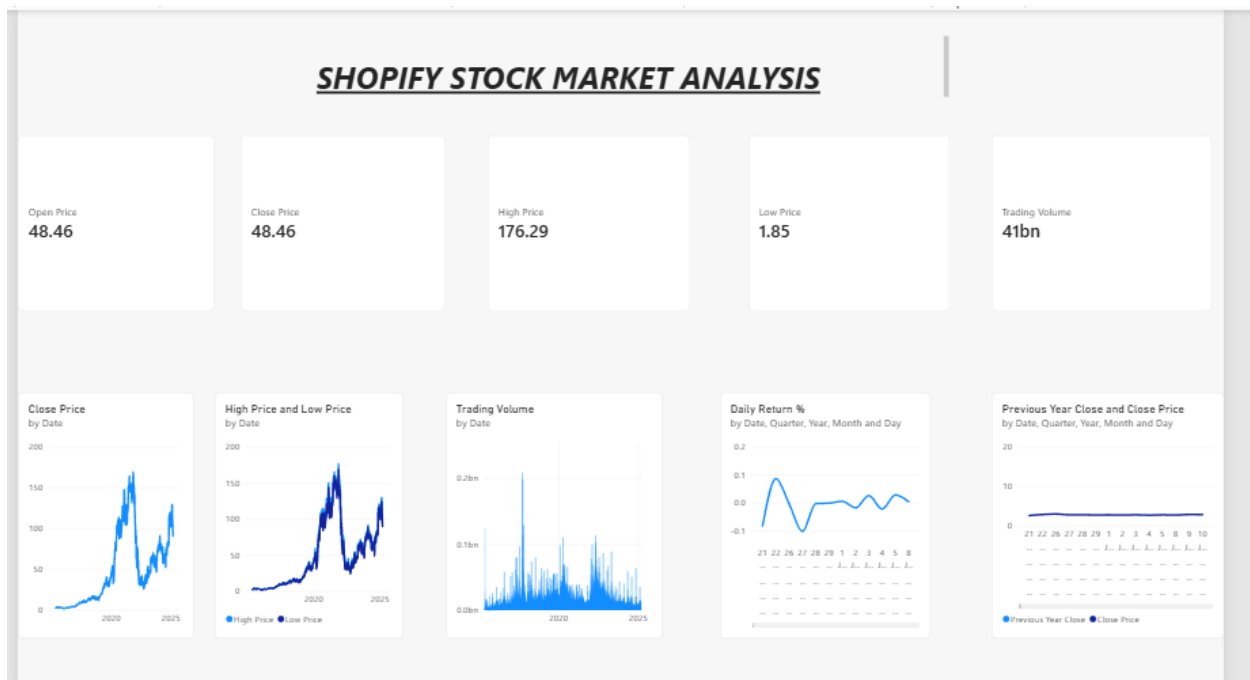
The dashboard also contains charts for stock price trends, trading volume, daily returns and previous-year comparison.

 **INSERT SCREENSHOT 5 HERE**

Use: Your **final dashboard screenshot** — the one showing:

- SHOPIFY STOCK MARKET ANALYSIS
- Open Price

- Close Price
- High Price
- Low Price
- Trading Volume
- Close Price chart
- High Price and Low Price chart
- Trading Volume chart
- Daily Return % chart
- Previous Year Close and Close Price chart



13. Final Deliverables

Requirement	Completed
Shopify stock data imported	✓
Stock data prepared	✓
Dim_Date created using CALENDAR()	✓
Date table marked as Date Table	✓
One-to-many relationship	✓
Year-Quarter-Month-Day hierarchy	✓
Open Price measure	✓
Close Price measure	✓
High Price measure	✓

Low Price measure	✓
Trading Volume measure	✓
Daily Return % measure	✓
DATEADD()	✓
TOTALYTD()	✓
SAMEPERIODLASTYEAR()	✓
Power BI dashboard	✓